INTRODUCTION TO MACHINE LEARNING · TEAM 8 · CRITICAL ANALYSIS + EMPIRICAL DEMONSTRATION

Where PILOT's guarantees stop

Adaptive LLM Routing under Budget Constraints [1]: two weaknesses, argued and measured

Thái Kiệt · Trần Lê Hải · 24KHMT1 · University of Science, VNU-HCM

W&B · 3,626 runs

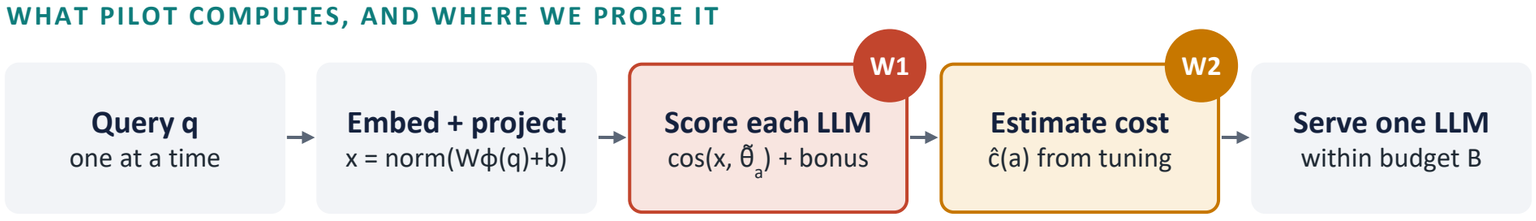
GitHub · code

Scan: every number's run, and the code

MAIN FINDING

PILOT's guarantees hold only inside their assumptions: **cosine scoring erased a 4× reward gap** (regret 600 vs 1.2), and **a 10% cost error pushed 23 of 48 runs over budget**.

WHAT PILOT COMPUTES, AND WHERE WE PROBE IT



Our reproduction, in short

Lowest regret among learners: **4,275 vs 4,747** (RouterBench)

50% allocation, quality / GPT-4: **PILOT 86.6% vs LinUCB 91.3%**

8 of 11 arms · text-embedding-3-small · 3 seeds

W1 Cosine scoring discards reward magnitude

THEORY · WHY IT SHOULD FAIL

Assumption. The PI-OFUL bound is proven for linear rewards $r = \theta^* \cdot x + \varepsilon$ [2]: the size of $\theta^* \cdot x$ is the reward.

What PILOT scores. $\cos(x, \tilde{\theta}_a) = x \cdot \tilde{\theta}_a \,/\, \|\tilde{\theta}_a\|$: scaling $\tilde{\theta}_a$ by any $c > 0$ changes nothing.

Consequence. Arms that point the same way but pay 0.2 and 0.8 get equal means. Only the exploration bonus separates them, so they are served alternately and regret grows like Θ(T).

Why it matters. LLMs often agree on which queries are hard (direction) but differ in how often they succeed (magnitude).

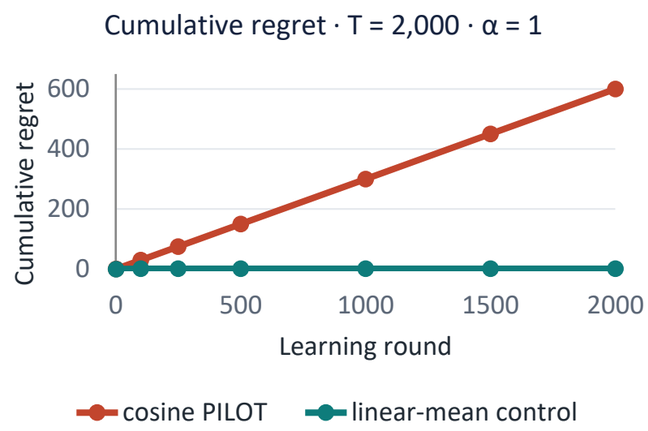
Prediction. Collinear arms: cosine regret grows ∝ T; a linear-mean control with the same prior stays near 0.

predicts

EXPERIMENT · DESIGNED FOR THIS WEAKNESS

x = (1, 0) every round · rewards 0.2 / 0.8 · oracle priors (0.2, 0), (0.8, 0) · λ = 1

Same A, b and selected-only updates. **Control**: linear mean $x \cdot \tilde{\theta}$ instead of cosine. Grid T × α = 3 × 3.



All 9 settings: cosine / control regret

T \ α	0.5	1	2
100	30/0.0	30/1.2	30/3.6
500	150/0.0	150/1.2	150/4.8
2,000	600/0.0	600/1.2	600/5.4

Cosine regret is exactly 0.3 × T in every setting: the tied arms are served alternately.

600 vs 1.2

regret after 2,000 rounds, α = 1 · synthetic

Verdict. The predicted magnitude blindness appears in all 9 settings. **Limits:** a deliberately assumption-violating synthetic case, independent of the embedder. It shows PILOT's score leaves the theorem's model; it is not a counterexample to the conditional OFUL theorem. The control shares PILOT's prior and is not the zero-prior LinUCB baseline.

W&B [3]: cosine pilot-5364678c… · control pilot-38312104… · all 18 runs in group lab2-critique

W2 Budget feasibility holds for estimated, not realized, cost

THEORY · WHY IT SHOULD FAIL

Guarantee. ZCL is competitive when each item's weight is known on arrival [4].

What PILOT knows. An estimate $\hat{c}_t(a)$ from input tokens and the tuning-mean output length. The bill is $c_t(a) = \hat{c}_t(a)(1 + e_t)$.

Consequence. Spend = $\Sigma\hat{c} + \Sigma\hat{c} \cdot e_t$; the policy controls only the first sum. A bin spent to its edge ($z \to 1$) turns any bias $e > 0$ into overspend.

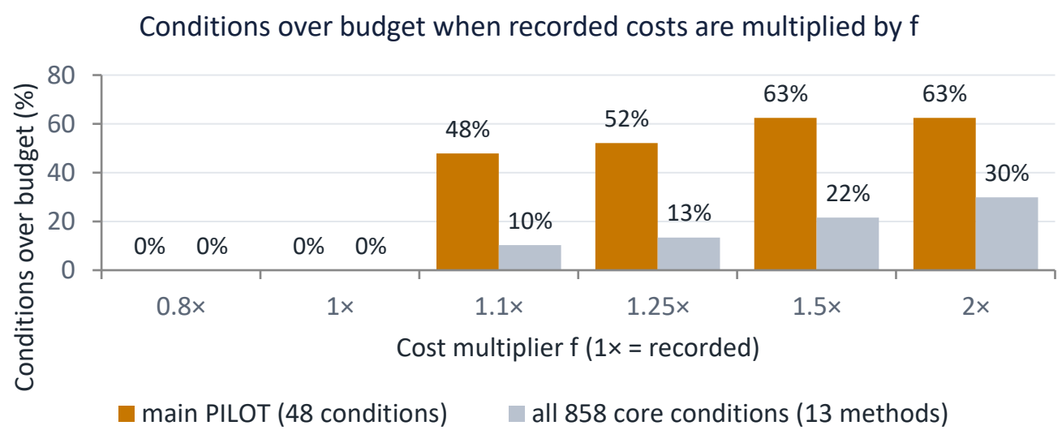
Hidden assumption. Deployment answers are as long as tuning answers, and prices hold.

Prediction. A few percent of cost error flips fully spent runs over budget, although every choice looked feasible.

predicts

EXPERIMENT · DESIGNED FOR THIS WEAKNESS

All **48** main-PILOT, full-learning conditions of the OpenAI study (2 populations × 8 allocations × 3 seeds). Multiply each chosen answer's recorded cost by **f**; keep choices and rewards fixed; count conditions whose spend exceeds the allocation.



0 → 23 / 48

conditions over budget at 1.0× → 1.1× recorded cost

Verdict. Feasibility is a property of the estimate, not of the bill. Main PILOT is more exposed than the average condition (48% vs 10% at 1.1×): it spends close to its allocation (median 86% vs 16%). Historical BGE run: 0 → 25 / 48. **Limits:** fixed-choice counterfactual; counts are conditions, not queries or probabilities; no provider was charged. At 1.0× none of the 858 core conditions overspends; 7 of 240 ablation conditions do.

W&B [3]: 1.0× p2-0071b43e796… · 1.1× p2-174179870de… · all core at 1.0× p2-b332a34c98e…

Three strengths, with evidence

S1 One reward per query. Only the served answer is rated, yet regret is lowest among learners (4,275 vs 4,747).

S2 Cold start from free data. A prior from public preference pairs: 60.7% calibration accuracy vs 50% chance.

S3 Auditable deployment. Frozen router + explicit cost policy: 0/48 recorded overspends.

Theory audit: more limits

Rate. The prior shrinks the $\sqrt{\lambda} \cdot S$ constant, not the $d\sqrt{T}$ rate.

S′ unknown. λ_a = 1/accuracy stands in for it; λ = 1 did better (+1.55 pp / +0.32 pp).

Proof. Two printing slips (pp. 23945–46). Corrected, the bound holds.

Cost. $O(K d^2)$ per query is possible with rank-one updates (ours: dense $O(K d^3)$); embedding dominates.

Future directions, one per weakness

F1 → W1. A magnitude-aware score (linear mean or learned arm scale) so the OFUL analysis applies. Test: magnitude sweep + RouterBench, same prior and budget.

F2 → W2. Reserve budget from tuning-only cost quantiles, updated from realized bills. Test: overspend and coverage under length shifts.

Both are proposals, not yet demonstrated.

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[2] Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári, "Improved algorithms for linear stochastic bandits," in Adv. Neural Inf. Process. Syst., vol. 24, 2011, pp. 2312–2320.
[3] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[4] Y. Zhou, D. Chakrabarty, and R. Lukose, "Budget constrained bidding in keyword auctions and online knapsack problems," in Proc. 17th Int. Conf. World Wide Web, 2008, pp. 1243–1244.
[5] Team 8, "pilot-llm-routing," GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing